EXAMPLE: CARTESIAN ROBOT POSITION

$$P = [3, 4, 7]^T$$

CARTESIAN COORDINATES.

$$P_x = 3 \quad P_y = 4 \quad P_z = 7.$$

$$\therefore \quad T_{CART} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

EXAMPLE: CYLINDRICAL ROBOT POSITION

$$P = [3, 4, 7]^T$$

$$\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} r C\alpha \\ r S\alpha \\ l \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

$$l = 7 \quad - (1)$$

$$r C\alpha = 3 \quad - (2)$$

$$r S\alpha = 4 \quad - (3)$$

(3)/(2)

$$\frac{\cancel{r} S\alpha}{\cancel{r} C\alpha} = \frac{4}{3}$$

$$\tan\alpha = \frac{4}{3}$$

$$\therefore \alpha = \underline{\underline{53.1^\circ}} \quad - (4)$$

SUB (4) INTO (2)

$$r\, C(53.1^\circ) = 3$$

$$\therefore r = 3/0.6 = 4.9965$$

$$r \approx \underline{\underline{5}}\, m$$

EXAMPLE: SPHERICAL POSITION

$$P = [3, 4, 7]^T$$

$$\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} r S\beta . C\gamma \\ r S\beta S\gamma \\ r C\beta \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 7 \end{bmatrix}$$

$$r S\beta C\gamma = 3 \quad -(1)$$

$$r S\beta S\gamma = 4 \quad -(2)$$

$$r C\beta = 7 \quad -(3)$$

(2)/(1)

$$\frac{\cancel{r} \cancel{S\beta} S\gamma}{\cancel{r} \cancel{S\beta} C\gamma} = \frac{4}{3}$$

$$\tan\gamma = 4/3.$$

$$\therefore \gamma = \underline{\underline{53.1^\circ}} \quad \text{OR} \quad \underline{\underline{233.1^\circ}}$$

SUB INTO (1).

$$r S\beta = \frac{3}{0.6} = 5m \quad \text{OR} \quad -5m \quad -(4)$$

(4)/(3)

$$\frac{r S\beta}{r C\beta} = \tan\beta = \frac{5}{7} \quad \text{OR} \quad \frac{-5}{7}$$

$$\therefore \beta = \underline{\underline{35.5^\circ}} \quad \text{OR} \quad \underline{\underline{-35.5^\circ}}$$

SUB INTO (4)

$$r = 5/S(35.5^\circ) \quad \text{OR} \quad -5/S(-35.5^\circ)$$

$$= \underline{\underline{8.33m}} \quad \text{OR} \quad \underline{\underline{8.33m}}$$